

Companion LXXXII: Transport-Based Parameter Inference and α -Sensitivity Maps for the Relaxation-Driven Cyclic Cosmology

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Abstract

Transport-based statistics provide a natural way to compare distributions of persistence diagrams across cosmological models. Building on the diffusion-optimal transport framework developed in previous Companions, this work introduces transport-based parameter inference and α -sensitivity maps for the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation parameter α controls the scale-dependent suppression of the gravitational potential, which in turn deforms the distribution of weak-lensing persistence diagrams. We construct transport distances and neural transport maps between RDCC and Λ CDM ensembles, and define sensitivity fields that quantify how these transport statistics respond to variations in α . The resulting framework provides a direct link between geometric transport quantities and observable constraints on α from Euclid-like weak-lensing surveys.

1 Introduction

Weak-lensing convergence maps encode the projected large-scale structure of the Universe. Their topological structure can be quantified using persistence diagrams, which capture the birth and death of connected components, loops, and voids. Previous Companions developed a hierarchy of geometric tools for these diagrams, including Wasserstein distances, entropic optimal transport, Schrödinger bridges, diffusion models, and neural transport flows.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation parameter α controls the scale-dependent suppression of the gravitational potential. This induces characteristic deformations in the distribution of persistence diagrams relative to Λ CDM. The goal of this Companion is to translate these deformations into *transport-based parameter constraints* and explicit *α -sensitivity maps* that can be evaluated on Euclid-like mock catalogues.

2 Transport Distances as Summary Statistics

Let $\mathcal{D}_\Lambda$ and $\mathcal{D}_\alpha$ denote ensembles of persistence diagrams from Λ CDM and RDCC with parameter α , respectively. We define a transport-based discrepancy

$$\mathcal{T}(\alpha) = \mathbb{E}_{D \sim \mathcal{D}_\Lambda} [W_p(D, T_\alpha(D))], \quad (1)$$

where W_p is a Wasserstein or Sinkhorn distance, and T_α is a neural transport map pushing $\mathcal{D}_\Lambda$ towards $\mathcal{D}_\alpha$.

The function $\mathcal{T}(\alpha)$ acts as a *summary statistic*: it vanishes when the two ensembles are indistinguishable and grows as RDCC deviates from Λ CDM. Its scale dependence can be probed by restricting diagrams to features in specific birth–death or angular multipole ranges.

3 α -Sensitivity Maps

We define the α -sensitivity of the transport statistic as

$$S_\alpha(\ell) = \left. \frac{\partial}{\partial \alpha} \mathcal{T}_\ell(\alpha) \right|_{\alpha=\alpha_0}, \quad (2)$$

where $\mathcal{T}_\ell(\alpha)$ is the transport discrepancy computed from features associated with angular scale ℓ , and α_0 is a fiducial value.

In practice, $S_\alpha(\ell)$ is estimated by finite differences using RDCC mocks at nearby α values. The resulting function provides an ℓ -resolved *sensitivity map* indicating which scales contribute most strongly to constraining α .

4 Likelihood-Free Inference with Transport Statistics

Given observed persistence diagrams $\mathcal{D}_{\text{obs}}$ from a Euclid-like survey, we define an empirical transport discrepancy

$$\hat{\mathcal{T}}(\alpha) = \mathbb{E}_{D \sim \mathcal{D}_{\text{obs}}} \left[W_p(D, T_\alpha^{\text{mock}}(D)) \right], \quad (3)$$

where T_α^{mock} is trained on simulations at parameter α . This quantity can be used in a likelihood-free inference scheme:

- sample α from a prior,
- generate RDCC mocks and train T_α^{mock} ,
- compute $\hat{\mathcal{T}}(\alpha)$,
- accept or weight α according to the discrepancy.

The resulting posterior on α is driven directly by transport-based topological differences between RDCC and Λ CDM.

5 Connection to Previous Geometric Companions

The present framework leverages several structures developed earlier:

- Wasserstein and Sinkhorn distances (Companions LXXVI–LXXVII),
- Schrödinger bridges and diffusion-regularized transport (LXXVIII),
- score-based diffusion models (LXXIX),
- neural Schrödinger flows and unified diffusion–OT geometry (LXXX–LXXXI).

Neural transport maps T_α can be instantiated as:

- the mean flow of a neural Schrödinger SDE,
- the deterministic component of a unified diffusion–OT model,
- or an explicit OT map parameterized by a neural network.

In all cases, the RDCC parameter α enters through the modified convergence field and the resulting deformation of the persistence-diagram distribution.

6 Conclusion

Transport-based statistics provide a natural and geometrically meaningful way to compare persistence-diagram ensembles across cosmological models. In the Relaxation-Driven Cyclic Cosmology, the relaxation parameter α induces scale-dependent deformations that can be captured by neural transport maps and Wasserstein or Sinkhorn discrepancies. The α -sensitivity maps introduced here quantify which angular scales and topological features contribute most strongly to constraining α in Euclid-like weak-lensing surveys. The present Companion thus connects the geometric machinery of previous Companions directly to observable parameter inference, addressing the need for explicit links between theory and constraints.

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